THE SATURATION NUMBER OF $K_3 \cup 2K_2$

Abstract. Let $F = K_3 \cup 2K_2$. Lin, He, and Xu recently proved that $\mathrm{sat}(n, F) = 15$, with unique extremal graph $K_6 \cup I_{n-6}$, for $n \geq 31$. We determine all remaining orders:

$$\mathrm{sat}(n, F) = \begin{cases} \binom{n}{2}, & 1 \leq n \leq 6, \\ n+2, & 7 \leq n \leq 12, \\ 15, & n \geq 13. \end{cases}$$

For every $n \geq 14$, the unique extremal graph is $K_6 \cup I_{n-6}$; the threshold is sharp because nonisomorphic extremal graphs occur at $n = 13$. Thus the $(p, q, r) = (2, 2, 3)$ instance of the three-clique saturation problem is settled for all orders. The finite verification exhausts all isolate-free graphs through 14 edges and all relevant 15-edge graphs of order at least 14.

1. Main result

All graphs are finite, simple, and undirected, and copies are not required to be induced. A graph G is F-saturated if it is F-free but $G + e$ contains a copy of F for every nonedge e of G. The saturation number $\mathrm{sat}(n, F)$ is the minimum number of edges in an n-vertex F-saturated graph. Write I_m for the edgeless graph on m vertices.

Currie, Faudree, Faudree, and Schmitt record the determination of $\mathrm{sat}(n, K_p \cup K_q \cup K_r)$ and $\mathrm{sat}(n, 2K_p \cup K_q)$ as Problem 3 of their survey [1]. Lin, He, and Xu prove that, when $2 \leq p \leq q < r < p + q$ and

$$n > 3(p-2) + (q+r)(q+r+1),$$

the saturation number and the unique extremal graph are determined [2]. For $(p, q, r) = (2, 2, 3)$, their theorem applies for $n \geq 31$ and gives the value 15 with extremal graph $K_6 \cup I_{n-6}$. We complete this parameter instance and determine the sharp uniqueness threshold.

Theorem 1. *For $F = K_3 \cup 2K_2$,*

$$\mathrm{sat}(n, F) = \begin{cases} \binom{n}{2}, & 1 \leq n \leq 6, \\ n+2, & 7 \leq n \leq 12, \\ 15, & n \geq 13. \end{cases}$$

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For every $n \geq 14$, the unique extremal graph up to isomorphism is $K_6 \cup I_{n-6}$. At $n = 13$ there are nonisomorphic extremal graphs, so the uniqueness threshold is sharp.

2. Constructions

Lemma 2. *For every $n \geq 7$,*

$$\mathrm{sat}(n, F) \leq \min\{n + 2, 15\}.$$

Proof. The graph $K_6 \cup I_{n-6}$ is F-free because it has only six nonisolated vertices. Every nonedge has an isolated endpoint; after adding it, use the new edge as one K_2 and choose a disjoint $K_3 \cup K_2$ among the remaining vertices of K_6. Hence $\mathrm{sat}(n, F) \leq 15$.

For $n \geq 8$, let B_n be obtained from a core K_4 on u_1, u_2, u_3, u_4 by attaching one leaf to each u_i and attaching the remaining $n - 8$ leaves to u_1. Then $e(B_n) = n+2$. Every triangle lies in the core, and deleting its vertices leaves a star and isolated vertices; hence B_n is F-free.

Let $s(x)$ denote the support of a leaf x. If the nonedge xu_i is added with $u_i \neq s(x)$, then $\{x, u_i, s(x)\}$ is a triangle, while pendant edges at the two remaining core vertices supply the two disjoint edges. If a nonedge xy between leaves with distinct supports is added, use xy as one edge, a pendant edge at a core vertex outside $\{s(x), s(y)\}$ as the other, and the other three core vertices as the triangle. If $s(x) = s(y)$, then $\{x, y, s(x)\}$ is a triangle and pendant edges at two other core vertices complete a copy of F. Thus B_n is F-saturated.

For $n = 7$, take vertices $A = \{a_1, a_2, a_3\}$, $X = \{x_1, x_2, x_3\}$, and s, with

$$E(G_7) = \{a_i a_j : 1 \leq i < j \leq 3\} \cup \{a_i x_i, s x_i : 1 \leq i \leq 3\}.$$

The only triangle is A, and $G_7 - A$ is a star, so G_7 is F-free. Its nonedges are $x_i x_j$, $s a_i$, and $a_i x_j$ for $i \neq j$. For $\{i, j, k\} = \{1, 2, 3\}$, adding $x_i x_j$ gives the triangle A and the disjoint edges $x_i x_j, s x_k$; adding $s a_i$ gives the triangle $\{s, x_i, a_i\}$ and the disjoint edges $a_j x_j, a_k x_k$; and adding $a_i x_j$ gives the triangle $\{a_i, a_j, x_j\}$ and the disjoint edges $s x_i, a_k x_k$. Hence G_7 is F-saturated and has 9 edges. ☐

3. Lower bound

We first remove graphs with isolated vertices from the finite search. We use the following immediate specialization of Proposition 3.4 of Lin, He, and Xu [2].

Lemma 3. *Let H be a nonempty graph such that*

1. *H contains no pairwise vertex-disjoint copies of K_2, K_2, K_3;*
2. *for every $z \in V(H)$, the graph $H - z$ contains vertex-disjoint copies of K_2 and K_3.*

Then H contains either two vertex-disjoint triangles or vertex-disjoint copies of K_2 and K_4.

Proof. If neither configuration occurred, then H would contain neither disjoint K_3, K_3 nor disjoint K_2, K_4. Together with the two stated hypotheses, these are precisely the assumptions of Proposition 3.4 of [2] with $(q, r) = (2, 3)$, which asserts that no such nonempty graph exists. □

Lemma 4. *Let $n \geq 7$. If an n-vertex F-saturated graph has an isolated vertex, then it is isomorphic to $K_6 \cup I_{n-6}$.*

Proof. Let I be the set of isolated vertices of an F-saturated graph G, and put $H = G - I$. Then H is nonempty and has no isolated vertices. Fix $v \in I$. For every $z \in V(H)$, the copy of F created by adding vz must use vz as one of its two K_2 components. Hence $H - z$ contains vertex-disjoint copies of K_2 and K_3. Since G is F-free, H contains no pairwise disjoint K_2, K_2, K_3, so Lemma 3 applies.

Suppose first that H contains disjoint triangles A and B. If $x \notin A \cup B$, choose an edge xy incident with x. If $y \in A$, then xy, the edge of A avoiding y, and the triangle B form F; the case $y \in B$ is symmetric. If $y \notin A \cup B$, use xy, any edge of B, and the triangle A. Thus no such x exists.

Now suppose that H contains a disjoint edge ab and a K_4 on a set B. Put $X = V(H) \setminus (\{a, b\} \cup B)$. The set X is independent, and there are no edges from X to B, since either type of edge together with ab and a suitable triangle of B would form F. As H has no isolated vertices, each $x \in X$ is adjacent to a or b, but not both: otherwise xab is a triangle disjoint from a matching of size two in B. If, say, $xa \in E(H)$, then x is isolated in $H - a$, whereas $H - a$ contains a disjoint $K_2 \cup K_3$; together with xa this forms F. Hence $X = \varnothing$.

In either case $|V(H)| = 6$. If H had a nonedge, adding it would leave every vertex of I isolated and therefore could not create the seven-vertex graph F. Thus $H = K_6$. □

The remaining statement is finite and was verified exactly.

Lemma 5 (Computer verification)**.** *Among isolate-free F-saturated graphs with at most 14 edges, the minimum edge counts for orders $7 \leq v \leq 12$ are*

v	7	8	9	10	11	12
$\min e$	9	10	11	12	13	14.

There is no isolate-free F-saturated graph with at most 14 edges and order at least 13, and there is no isolate-free F-saturated graph with 15 edges and order at least 14.

Proof. For each $1 \leq e \leq 14$, we processed McKay's complete `graph6` catalogue of nonisomorphic simple graphs with exactly e edges and no isolated vertices [3, 4]. These catalogues contain 1,008,629 graphs in total.

The verifier uses the following exact criterion. A graph C is F-free if and only if

$$\nu(C - T) \leq 1 \quad \text{for every triangle } T \subseteq C,$$

where ν denotes matching number. Once this condition is checked, a nonedge xy is saturating if and only if either

(1) $C - \{x, y\}$ contains vertex-disjoint copies of K_3 and K_2, or
(2) some common neighbor z of x and y satisfies $\nu(C - \{x, y, z\}) \geq 2$.

Indeed, every new copy of F must use xy, and these alternatives correspond to xy being a K_2 component or lying in the triangle. Exhausting all nonedges gives the table and excludes every order at least 13 through 14 edges.

For the 15-edge layer, delete an edge from an isolate-free graph and then remove any endpoints that become isolated. The resulting isolate-free graph has 14 edges. Reversing this operation shows that every isolate-free 15-edge graph is obtained from an isolate-free 14-edge graph by adding an edge between two existing nonadjacent vertices, attaching a new leaf, or adjoining a disjoint K_2. Applying these extensions, subject to resulting order at least 14, to the 740,226 isolate-free 14-edge representatives produces 25,937,431 candidates, with repetitions allowed. None is F-saturated. □

Proof of Theorem 1. If $n \leq 6$, adding one edge cannot create the seven-vertex graph F, so the only F-saturated graph is K_n.

Let $n \geq 7$. By Lemma 2, an extremal graph G has at most 15 edges. If G has an isolated vertex, Lemma 4 gives $G \cong K_6 \cup I_{n-6}$ and $e(G) = 15$. Otherwise $\delta(G) \geq 1$, so $n \leq 2e(G) \leq 30$, and Lemma 5, together with Lemma 2, gives the asserted values.

For $n \geq 14$, Lemma 5 excludes an isolate-free extremal graph, and Lemma 4 gives uniqueness. At $n = 13$, the graphs B_{13} and $K_6 \cup I_7$ are nonisomorphic and extremal. □

Exact verification. The finite part of the proof is recomputable from the data printed here. One reads McKay's catalogues of nonisomorphic isolate-free graphs with exactly e edges for $1 \leq e \leq 14$ [3], 1,008,629 graphs in all, of which 740,226 have 14 edges; applies the F-freeness and saturation criteria displayed in the proof of Lemma 5, which use nothing beyond triangle enumeration and matching number; and expands the 14-edge layer by the three extensions listed there to the 25,937,431 candidates of order at least 14. This returns the minimum edge counts $9, 10, 11, 12, 13, 14$ for $7 \leq v \leq 12$, no F-saturated graph of order at least 13 with at most 14 edges, and none of order at least 14 with 15 edges. Every step is exact: all arithmetic is on integers, no floating-point computation is involved, and no isomorphism test beyond the published catalogues is used. The program that carried out this census, and its transcript, are not included with this note; they are retained by the authors and are available on request. The program and transcript that do accompany this note are those of a second, narrower verification: it reads no catalogue file, regenerating instead every isolate-free F-free graph of order at most 13 with at most 14 edges, which suffices for $\mathrm{sat}(n, F)$ with $n \leq 13$, and recomputing the three enumeration counts above by Pólya counting; it does not revisit the 14-edge layer in orders 14 through 28, nor the 15-edge layer.

References

[1] B. L. Currie, J. R. Faudree, R. J. Faudree, and J. R. Schmitt, *A survey of minimum saturated graphs*, Electron. J. Combin., Dynamic Survey DS19, second edition, 2021, doi:10.37236/41.

[2] H. Lin, Z. He, and Y. Xu, *A note on the saturation number for unions of three cliques*, arXiv:2608.00459v1 [math.CO], 2026, doi:10.48550/arXiv.2608.00459.

[3] B. D. McKay, *Simple graphs by number of edges*, Combinatorial Data, `https://users.cecs.anu.edu.au/~bdm/data/graphs.html`, accessed 23 August 2026.

[4] B. D. McKay and A. Piperno, *Practical graph isomorphism, II*, J. Symbolic Comput. **60** (2014), 94–112, doi:10.1016/j.jsc.2013.09.003.